UPS—Home

Ready Reckoner

Mid-end **You Need**	To provide power to a low or medium configuration consumer PC
Look for	An offline or line-interactive UPS with a 450 VA to 650 VArating, and a backup time of at least 10 minutes
High-end **You Need**	To provide power to a PC using high-end components, or a server
Look for	1 KVA rating, 10-minute backup, support for remote monitoring, alarms for power anomalies

Future

The next trend in the Indian UPS industry should see a major shift towards reliable and intelligent products. UPSes are as important for mission-critical applications as any other element, and reliability is a major issue.

At the same time, intelligent products will be able to maximise power availability for the equipment that needs it most, shutting down non-necessary equipment at different stages of battery use. Running costs will also be a driving factor.

Do Remember

- Decide on a UPS based on the total power consumed by your equipment. If it's expressed in amps, multiply that by 230 V—the nominal line voltage. If the power drawn is stated in watts, multiply it by a factor of 1.3 to 1.4 for the VA load. For example, a 250-watt power supply (x 1.4) equals a 350 VA load. Considering a measure of safety, 500 VA is enough for most PCs.

- The VA rating of a UPS is a measure of the amount of power it can provide—usually for about 10 to 12 minutes. Therefore, if a 500 VA UPS is loaded with equipment that draws 500 VA of power, it will last for about 10 minutes. If the load is halved to 250 VA, the backup time will increase to about 16 to 18 minutes. Of course, this is assuming that the battery is fully charged and functional.
- It is advisable to charge and discharge the UPS battery completely once a month to prolong battery life.
- A typical PC power supply is rated at 300 W which is the peak that it can deliver, however under normal load it is seldom reached, hence a 500VA UPS is enough for normal users. Gamers who use higher capacity power supplies may look for 700 VA and above UPS.
- All UPS vendors have good service network and it is just a call away. Make sure you have their direct contact number, so in case of emergency you can call them up for quick replacement of dead batteries etc.

3. One very important feature that you should lookout in an UPS is AVR (Automatic Voltage Regulation). This circuit helps in conditioning the incoming power supply and hence protects the PC against voltage fluctuations.

- Check out the software that UPS vendors provide. Some software are smart enough to save your current work and then proceed to a proper shutdown on the PC.
- If you have a small office with no more than five PCs it is advisable to go for a single UPS rather than one for each PC. You should contact your electrician for the necessary procedures.

Buying Tips

- INPUT VOLTAGE RANGE: This is the range beyond which the UPS delivers power through its battery. The greater the range the UPS can handle, the more versatile its power protection. Look for a voltage range between 150 and 270 V.
- VA-RATING: This is the power rating that can be provided by the UPS. As a rule of thumb, a 1 GHz processor-based system with a 20 GB hard disk, a CD-ROM drive and a 17-inch colour monitor would require a 500 VA UPS.
- BACKUP TIME: This is the specified time for which the UPS can provide power from its battery for a given load. Look for a backup time of at least 10 minutes at maximum load.
- SWITCHING OR TRANSFER TIME: This is the time taken by the UPS to switch from normal delivery power mode to battery mode on detecting a power failure. Ideally, look for transfer times under 5 milliseconds.
- ALARMS AND CONNECTIVITY: Most UPS systems have front panel LEDs that indicate the battery level and the condition of the input and output power. This comes in useful especially during power failures.
- CHARGING TIME: This is the time taken to charge the UPS batteries from the completely drained state to the fully charged condition. The charging time should be between 4 and 6 hours.